

5CS045 Student Server: What We Built and How We Run It

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In short

The 5CS045 server gives every Full Stack Development student their own login, three websites and a database. It runs on one Ubuntu 24.04 virtual machine hosted by the college IT team, and everything inside it is set up and run by us. If IT gives it all the RAM and CPU for exams, that is manageable from our side: the extra power is used straight away, and we raise a few limits so the database and web server use it too (see Exam time).

What	Address
Sign-up page and phpMyAdmin	https://fullstack.heraldcollege.edu.np/
Students' websites	https://fullstack-student.heraldcollege.edu.np/~username/
Admin panel (staff only)	https://fullstack.heraldcollege.edu.np:8443
SSH, SCP and SFTP	fullstack.heraldcollege.edu.np , port 50222
The VM itself	10.80.0.250

What the VM has

The VM has 8 vCPUs, 11 GB of memory and a 443 GB disk, and it is nearly idle today. Every student has fixed caps on files, database size, SSH use and website processes, so no single student can fill the disk or the memory. Their website and database use are capped more loosely: one student's pages can keep up to 4 cores busy.

Resource	Allocated	In use on 30 Sep
Processor	8 vCPUs	Load 0.06 (almost nothing)
Memory	11 GB	10 GB still free

Resource	Allocated	In use on 30 Sep
Disk	443 GB	15 GB used (4%)
Students	Room for about 670 at their full allowance	8 registered

Limits on each student

Limit	Per student	What happens at the limit
Files	500 MB	"Disk quota exceeded". We can raise it for one student on the panel.
Database	100 MB	It turns read-only until they delete data (checked every 15 minutes).
Processor, over SSH	1 core	They slow down; nobody else does.
Memory, over SSH	1 GB	Their programs slow down, then the biggest one is stopped.
Programs running, over SSH	200 (a backup cap of 300)	New ones are refused, so a fork bomb cannot take the server down.
Website (PHP)	4 pages at once, 128 MB and 30 seconds each	Extra visitors wait their turn; a page over 128 MB or 30 seconds fails.
Upload through a web form	20 MB	The upload is refused.

Built for a whole lab logging in at once

- SSH lets up to 1,000 people type their passwords at the same time (the default starts refusing at 10).
- The system bus takes 4,096 connections at once (the default is 256), so a class arriving over a minute or two all get their limits.
- PHP can keep 65,536 files open, enough for 800 or more student websites.
- An SSH connection whose computer has gone away (laptop shut, network lost) is closed after 10 minutes, and a password must be typed within 60 seconds.

What each student gets

Each student gets one login, three websites and a database, and signs themselves up. An account only ever exists for a real college mailbox.

- **One login** for SSH, SCP, SFTP and phpMyAdmin. The username comes from the email: sarayu.gautam@heraldcollege.edu.np becomes sarayu_gautam.
- **Three websites** at <https://fullstack-student.heraldcollege.edu.np/~username/> followed by workshops/, exam/ or assessment/.
- **A database** with the same name as their username.
- **Their own copy of PHP.** Their pages run as them, not as a shared web user, so one student's code cannot touch another's.
- **The tools the module uses:** PHP 8.3, MariaDB (MySQL), Git, Composer and npm. The network blocks packagist.org, so students run composer install on their laptop and upload the vendor folder.

How a student signs up

1. They open the sign-up page and type their college email.
2. The server checks it: only @heraldcollege.edu.np, nothing added to it (no name+1@), and not already registered.
3. It emails them a one-time link that works for 1 hour. Nothing is created yet.
4. They open the link and click **Create my account**.
5. The account is made and their username and password arrive in a second email. Their websites work about 10 seconds later.

A made-up or mistyped address never gets past step 3, and each email can sign up only once. Students cannot reset their own password; a tutor does that on the admin panel, where tutors can also add one student or a whole class from a list of emails.

Security

Students are walled off from each other and from the server, logins are guarded against password guessing, and staff tools sit on a separate port that only listed computers can open. Outside testers (VAPT) are checking all of this now.

Students kept apart

- Each student's files are private. The web server can only read their three website folders.

- Their PHP runs as them, so it can reach only what their own login can.
- Their database login works on their own database only.
- They cannot use sudo, schedule jobs (cron and at) or install software for everyone (npm install -g is refused).
- Code inside npm and Composer packages runs with the rights of the student who installed it, nothing more.
- The copy of each password kept for tutors is readable by root only, never stored in the student's folder.
- The limits in "What the VM has" stop one student from starving the rest.

Logins

- SSH, SCP and SFTP use port 50222.
- 5 wrong passwords within 10 minutes block that computer for 1 hour; blocked 3 times in a day means a week. A tutor can unblock a lab on the panel.
- root cannot log in over SSH, and each connection gets at most 4 password tries.
- SSH's own forwarding and tunnels are switched off, though students' shells can still reach the college network (see Still open).

Websites

- Every page is HTTPS, and plain http:// is redirected to it.
- Students' websites have their own name, fullstack-student, apart from sign-up, phpMyAdmin and the panel, so a student's page can never read those logins.
- The admin panel does not answer on the students' name at all.
- Hidden files such as .env and .git are never served, PHP errors are logged instead of shown, and the email password is kept outside the website folders.

Sign-up

- Only @heraldcollege.edu.np addresses, only plain ones (no name+1@), and one account per email.
- Nothing is created until the emailed link is used. Links work once, for 1 hour, and are stored only as a fingerprint (a hash).
- The link's address comes from our settings, so a forged request cannot send students to another site.
- One link per email every 5 minutes, and 40 attempts per network every 15 minutes.
- The form is protected against forged submissions (CSRF), and every sign-up is logged.

Admin panel

- It has its own port, 8443, and nothing students see links to it.
- Only computers on its allowed list, and the server itself, can open it; everyone else gets "403 Forbidden" before any code runs. Students have a shell on the server, so they can reach the sign-in page, and the password is what stops them.
- Signing in needs a server account in the sudo group, or in the 5cs045-panel group for people who must not have sudo (such as testers). A panel account can still do everything on the panel, including showing passwords and removing students.
- 5 wrong passwords make that account and that computer wait 15 minutes, so a student guessing from the server could lock an admin out for that long. A session ends after 30 minutes idle, or 12 hours at most.
- The panel runs as its own user and can only ask the server for a fixed list of actions, each checked against a signed-in session. The public sign-up site cannot ask at all.
- Every change, sign-in and password view is logged with who did it and from which computer, and the pages block framing, outside scripts and caching.

Testing

- The health check runs about 50 checks of the above on the live server (panel: Semester, then Run the health check).
- The VAPT testers have 2 student accounts and a panel account without sudo. Testers are never given sudo.
- Load tests with 800 test accounts on 23 and 24 Sep found five problems. Four are fixed and the fifth now recovers by itself.

Still open

- The VM has no firewall of its own. From their shell, students can reach other machines on the college network and could start their own services on other ports; today only the college network limits this. A firewall on the VM that accepts only ports 80, 443, 8443 and 50222, and stops students' programs connecting to internal addresses or the panel, would close it.
- The sign-up site and phpMyAdmin run as one web user that may run the add-student script as root, and that script resets the password of any account that already exists, the admin's included. A break-in to either could take over student accounts or lock us out, so the script should refuse existing accounts when the website calls it.
- The certificate is self-signed, so browsers warn once. A trusted certificate, for example from Let's Encrypt, would remove the warning; the setup already has room for one.

- The testers' PDF, with their passwords, was committed to the repository on 30 Sep. Deleting the file does not remove it from the history, so change those passwords once testing ends, or straight away if the repository is public.
- Backups are with IT. We have asked for daily backups and a snapshot before and after each exam; this still needs their confirmation.

Running it day to day

Almost everything is done in a browser on the admin panel. SSH is needed only to install updates, change the email settings, or add a computer to the panel's list from that computer.

I want to	Do this
Help a student who cannot log in	Students, type their name, Reset password (tick the box to email it)
Resend a student's login email	Students, their account, Email their login
Unblock a lab after wrong passwords	Students, Unblock at the top of the page
Add one student, or a whole class	Students, Add; or Semester, add a class from a list of emails
Give a student more disk space	Students, their account, Disk limit
Check the server is healthy	Server page; or Semester, Run the health check
Let another staff computer open the panel	Security, allowed computers
Test the email settings	Security, Send a test email
See what happened	Logs: sign-ups, account changes and every panel action
Clear out at the end of term	Semester, remove every student
Install the latest version	SSH in, then <code>cd /opt/5cs045-provisioning && git pull && sudo ./setup/update-server.sh</code>

What runs by itself

- Every 15 minutes: databases over 100 MB are made read-only, oversized PHP error logs are trimmed, and an alert is emailed when the disk is 80% full.
- Every minute: the login service is restarted if a login rush has made it stop answering.
- After a reboot, the web server, PHP, database, SSH and fail2ban all start on their own.

How changes reach the server. All code lives in the GitHub repository

SarayuGautam/5cs045-student-provisioning, and every change goes there first. The server pulls it with a read-only deploy key over port 443, and the update script installs it without touching the email settings or any student.

Exam time: if IT gives the VM all the RAM and CPU

Yes, it is manageable from our end. The server uses extra CPU and memory as soon as the VM has them; we raise three settings once they arrive, and lower them again before IT takes the extra back.

What happens by itself

- Linux uses every CPU and all the memory it is given, and spreads logins, uploads and students' pages across them.
- Each student's share stays the same (1 core and 1 GB over SSH, 4 PHP pages at once). More students fit at once, and the fair split still holds.
- Nothing in our setup is tied to 8 vCPUs or 11 GB. The disk and network need no change.

What we change

Setting	Now	For exams
PHP processes for phpMyAdmin and sign-up	5	40
Database connections	151	500
Database cache	128 MB	4 GB

All three can be changed while the server is running, with no downtime. Without them, phpMyAdmin and the database would still queue students even on a much bigger VM.

What to agree with IT

1. Do the change before exam week, never during an exam. Adding memory or CPUs usually restarts the VM for a few minutes, unless IT has hot add switched on; everything starts again by itself.

2. Ask for the resources to be reserved, not just allocated: at least 32 GB and 16 vCPUs. "All" of a host that other VMs still share can leave an exam short at the busiest moment.
3. Once it is in, we check the Server page shows the new cores and memory, raise the three settings and run the health check.
4. On the day, lab finish times are staggered by 2 to 3 minutes, so a whole cohort never logs in within the same second.
5. After the exam we lower the three settings first; then IT takes the extra back, which also needs a restart.

Today the three settings are changed by hand. A small script could size them to whatever the VM has at every start, so a resize would need nothing from us.

What we did, newest first

The server went from its first test to today's setup in under two weeks, 18 to 30 Sep 2026. Code changes are tracked in the repository; SSH on port 50222, disk quotas, npm and the deploy key were set up by hand on the VM, so a rebuild must repeat them.

Date	What we did	Why
30 Sep	Sign-up creates the account only after the student opens a link emailed to them	Made-up or mistyped addresses were getting accounts
30 Sep	Sign-up refuses name+1@ addresses	One student could make many accounts, all mailed to the same inbox
30 Sep	Students' websites moved to their own name, fullstack-student; the panel refuses that name	Student code must never share an address with anything that has a login
30 Sep	Password-guessing protection moved to port 50222	It only guarded port 22, so guessing on 50222 carried on during a ban
30 Sep	Pages uploaded from Windows are made readable automatically	Windows' built-in scp gave "403 Forbidden"
29 to 30 Sep	Student guide written, then cut to 5 pages; web uploads raised to 20 MB	Students need one short guide; projects need bigger uploads

Date	What we did	Why
24 Sep	Admin panel on port 8443, and a panel group without sudo	Run everything from a browser; testers never get sudo
24 Sep	Login service restarts itself when stuck; no background process per student login	800 logins at once stalled the server
23 Sep	Load tests with up to 800 accounts; raised the SSH, login-session and PHP file limits	A whole lab could not log in at once
23 Sep	Per-student CPU, memory, process, disk and database limits	One student must not starve the rest
23 Sep	Week-long ban for computers that keep guessing passwords; sign-up limited per network	Bots waiting out the hourly ban, and sign-up abuse
23 Sep	Both names mapped; sign-up moved to the site's front page; phpMyAdmin's extra password prompt removed	Simpler for students
22 Sep	Saved passwords moved out of students' folders to a root-only place	Students could read or edit them
21 Sep	Folders named workshops, exam and assessment; database named after the username; welcome pages; email settings kept out of Git	A clear, consistent layout, and no secrets in the code
18 to 20 Sep	First version: sign-up page, account scripts, per-student PHP, websites and database; SSH with no root login or tunnels, and fail2ban	Start of the project

Documents that go with it

- The student guide (5 pages): docs/Server_Access_Guide.docx in the repository.
- The full admin manual: README.md in the repository.
- The VAPT test sheet: accounts and what to try, for the outside testers.
- The capacity brief for IT: [5CS045 Server Capacity for Exams](#), with the memory and CPU to ask for.